Tableau Interview Question Bank

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Basics, components & connections

Q1. What are the Tableau components? [Easy]

- **Tableau Desktop:** The primary application for creating and developing interactive visualizations, dashboards, and stories.
- **Tableau Server:** A platform for publishing and sharing interactive dashboards and workbooks with a wider audience within an organization. It enables collaboration, data governance, and scheduled refreshes.
- **Tableau Online:** A cloud-based version of Tableau Server, hosted and managed by Tableau. It offers similar functionalities to Tableau Server but eliminates the need for on-premises infrastructure.
- **Tableau Reader:** A free application that allows users to view and interact with dashboards and workbooks created in Tableau Desktop or published to Tableau Server/Online. It cannot edit or create new visualizations.
- **Tableau Public:** A free cloud-based platform for creating and sharing interactive visualizations with the public.

Q2. What is Tableau Desktop? [Easy]

Tableau Desktop is the core application for creating and developing interactive visualizations, dashboards, and stories. It allows users to connect to various data sources, create a wide range of visualizations, build interactive dashboards, and develop stories to guide users through a series of visualizations.

Q3. What is Tableau Server? [Easy]

Tableau Server is a platform for publishing and sharing interactive dashboards and workbooks with a wider audience within an organization. It enables collaboration, data governance, and scheduled refreshes. Tableau Server can be deployed on-premises or as a cloud-based service (Tableau Online).

Q4. What is Tableau Online? [Easy]

Tableau Online is a cloud-based version of Tableau Server, hosted and managed by Tableau. It offers similar functionalities to Tableau Server but eliminates the need for on-premises infrastructure, making it easier to deploy and manage.

Q5. What is Tableau Reader? [Easy]

Tableau Reader is a free application that allows users to view and interact with dashboards and workbooks created in Tableau Desktop or published to Tableau Server/Online. It cannot edit or create new visualizations, making it ideal for users who only need to view and consume data visualizations.

Q6. Difference between Tableau Desktop and Server? [Easy]

– Tableau Desktop | Tableau Server
– Authoring tool — create and develop visualizations, dashboards, stories | Sharing platform — publish, distribute and govern content
– Installed on the analyst's machine | Deployed on company infrastructure (or cloud)
– Connects to data and builds workbooks | Handles permissions, scheduled refreshes, collaboration

Q7. What databases are currently used in projects? [Easy]

The choice of database depends on the specific project requirements and the nature of the data. Some commonly used databases include:

- **Relational Databases:** SQL Server, Oracle, MySQL, PostgreSQL
- **Data Warehouses:** Snowflake, Amazon Redshift, Google BigQuery
- **Cloud Databases:** AWS RDS, Azure SQL Database, Google Cloud SQL
- **Other:** Excel, CSV files, JSON, etc.

Q8. What is the SQL / Oracle database connection process in Tableau? [Easy]

Establish Connection:

- Open Tableau Desktop.
- Go to "Connect" → "To Server" → "More…"
- Select the appropriate database driver (e.g., "SQL Server," "Oracle").
- Enter the server name, database name, and credentials.

Connect and Explore:

- Test the connection to ensure it's successful.
- Browse available tables and views within the database.
- Drag and drop fields onto the canvas to create visualizations.

Q9. What are the connection types? (Live & Extract) *[Easy]*

- **Live Connection:** Connects directly to the live data source, providing real-time data updates but requiring the data source to be available and accessible.
- **Extract:** Creates a local copy of the data, offering faster performance for large datasets but requiring manual refreshes for data updates.

Q10. How do you load (import) the database into Tableau? *[Easy]*

- **Connect:** Establish a connection to the data source.
- **Select Data:** Choose the tables or views you want to analyze.
- **Create Extract (Optional):** If necessary, create an extract for improved performance.
- **Start Analysis:** Begin exploring and visualizing the data.

Q11. What is custom SQL usage? Can you work with parameters? *[Medium]*

Custom SQL allows you to write your own SQL queries to extract specific data from the database. You can integrate parameters into your custom SQL queries to allow users to filter data dynamically.

Q12. Difference between VizQL and Custom SQL? *[Medium]*

- **VizQL:** Tableau's own visual query language — automatically generated when you drag and drop fields; it translates your visual actions into optimized queries.
- **Custom SQL:** hand-written SQL you provide at the data-source level to shape exactly what data enters Tableau, useful for complex joins/derivations the drag-and-drop interface can't express.

Q13. Difference between TWB and TWBX? *[Easy]*

- **TWB:** Tableau Workbook file containing the workbook definition and connections to external data sources.
- **TWBX:** Packaged Tableau Workbook file that includes the workbook definition and the underlying data itself, making it easier to share and distribute.

Q14. Which file type is used when working with Tableau Reader? *[Easy]*

TWBX files are used with Tableau Reader.

Q15. What is TDE and how does it improve performance? *[Medium]*

TDE (Tableau Data Extract) is a proprietary file format specifically designed for Tableau. It improves performance by optimizing data storage and retrieval, and allows for extracting only the necessary data for analysis. (Newer versions use the .hyper format.)

Performance, charts & data concepts

Q16. Ways to increase the performance of a dashboard/workbook/database? *[Medium]*

- **Data Extraction:** Create extracts for faster performance, especially with large datasets.
- **Data Subsetting:** Extract only the necessary data to reduce the size of the extract.
- **Data Aggregation:** Aggregate data at the source level to reduce the amount of data transferred.
- **Optimize Visualizations:** Use efficient visualization types and avoid overly complex calculations.
- **Server Optimization:** Tune server settings for optimal performance (if using Tableau Server).
- **Data Source Optimization:** Ensure the data source is well-indexed and optimized for querying.

Q17. How do you check performance recordings in Tableau Desktop? [Medium]

Enable performance recording in Tableau Desktop (Help → Settings and Performance → Start Performance Recording) and analyze the recorded information to identify performance bottlenecks.

Q18. Difference between Bar chart and Pie chart? [Easy]

- **Bar Chart:** Compares values across different categories, useful for showing trends, distributions, and comparisons.
- **Pie Chart:** Shows the proportion of each category to the whole, best for visualizing parts of a whole.

Q19. Explain Heat Map and Dual Axis [Easy]

- **Heat Map:** Uses color to represent the intensity of data values across a matrix, helping to identify patterns, trends, and anomalies in data.
- **Dual Axis:** Allows you to plot two different measures on the same chart using separate axes, enabling comparisons of related measures with different scales.

Q20. Example for Dual Axis and when to use synchronized axes? [Medium]

- **Example:** Plot sales and profit on the same chart.
- **Synchronized Axes:** Useful when comparing two related measures with different scales, ensuring that the axes are aligned and move together.

Q21. What is a Waterfall chart, and blended vs individual axes? [Medium]

- **Waterfall Chart:** Visualizes how an initial value is affected by a series of positive and negative increments.
- **Blended Axes:** Combine data from different sources on a single axis.
- **Individual Axes:** Use separate axes for each data source.

Q22. What are Dimensions and Facts? [Easy]

In Tableau, dimensions and facts are fundamental concepts used for data analysis and visualization.

Dimensions: These are descriptive attributes that categorize your data. They act like labels on chart axes, helping you understand the context of your measures. Examples: customer name, product category, region, date (year, quarter, month), employee department.

Facts: These are quantitative measures associated with your dimensions. They represent the numerical values you want to analyze and visualize. Examples: sales amount, units sold, profit margin, customer lifetime value, number of website visits.

Think of dimensions as providing the "who, what, when, and where" of your data, while facts represent the "how much" or "how many." By combining dimensions and facts in Tableau, you can create insightful visualizations that reveal patterns and trends in your data.

Q23. Can you describe a complex dashboard you've created or worked on? [Medium]

Yes, sure — I once built a dashboard for a retail company that monitored key performance indicators (KPIs) across various departments and sales channels. It was intricate because it involved:

- **Multiple Data Sources:** It combined data from sales transactions, customer relationship management (CRM), and inventory management systems.
- **Complex Calculations:** I created calculated fields to derive metrics like average order value, customer lifetime value, and inventory turnover.
- **Interactive Features:** The dashboard included filters, drill-downs, and parameter controls to allow users to explore the data by specific product categories, regions, or timeframes.
- **Advanced Visualizations:** I used a combination of charts (bar charts, line charts, heatmaps) and maps to present the data in an informative and visually appealing way.

Q24. Why is it complicated? (follow-up to your dashboard answer) [Medium]

Possible reasons for complexity:

- Multiple data sources requiring data blending and transformation.
- Incorporating complex calculations and custom formulas.
- Designing interactive elements for user exploration and analysis.
- Combining various chart types to effectively represent different data aspects.

Q25. What types of data sources have you used to create reports and dashboards? [Easy]

- **Relational Databases (RDBMS):** SQL Server, Oracle, MySQL, PostgreSQL (common for storing business data)
- **Spreadsheets:** Excel, CSV (often used for smaller datasets or initial data exploration)
- **Cloud Data Sources:** Salesforce, Google Analytics, social media APIs (provide access to web application or platform data)
- **Flat Files:** Text files, JSON (can be used for specific data formats or custom data feeds)

Q26–27. Have you created dashboards from a data warehouse? Which RDBMS have you worked with? [Easy]

Answer honestly based on your experience. If yes, briefly mention the benefits of using a data warehouse: a centralized, integrated data source optimized for analytics. Then list the specific RDBMS you're familiar with (e.g., SQL Server, Oracle, MySQL).

Q28. What is PK, FK and a composite key? [Easy]

Primary Key (PK): A unique identifier for a table row. Each row must have a distinct PK value, ensuring no duplicate records exist.

Foreign Key (FK): A column that references the PK of another table. This creates a relationship between tables, helping to maintain data consistency.

Composite Key: A combination of two or more columns that uniquely identifies a row in a table. It's used when a single column cannot guarantee uniqueness.

Tableau Server & advanced scenarios

Q29. Can I archive my most valuable workbooks and have version control at the same time? [Hard]

Tableau Server doesn't have a built-in archive feature. However, you can use:

- **Subscriptions:** Set up subscriptions to be notified when workbooks are updated. You can then export the workbook at that point for archival purposes.
- **Scheduling:** Schedule Tableau Server to export workbooks periodically for archiving.
- **Third-party tools:** Explore tools that integrate with Tableau Server to automate workbook backups and version control.
- **Version Control:** You can achieve a basic form of version control by saving workbooks as .twbx packages at different points in time. However, this requires manual intervention.

Q30. Can I be notified immediately when my extract fails (or succeeds)? [Medium]

Tableau Server can send email notifications for extract refresh failures or successes. You can configure this in the **Settings** → **Jobs** → **Email Notifications** section. Or you can use the Tableau API with Python integration to send a message on your phone.

Q31. Can I remove older workbooks (and archive them) from Tableau Server and simultaneously email the owner? [Hard]

- Tableau Server doesn't offer built-in functionality to remove workbooks and email owners simultaneously. This might require scripting or custom solutions.
- You can manually remove the workbook and then email the owner.
- Consider using the Tableau Server REST API to automate the removal process and potentially trigger emails through custom scripts.

Q32. What calculations (or fields) are in my Tableau workbooks? How do I grab the actual formula being used? [Medium]

In Tableau Desktop, open the workbook and go to the **Data** pane. Click on the dropdown arrow next to the field name and select **Show Formula** (or Edit). This will display the formula used to create the calculated field.

Q33. For HA and Tableau Server, how do I ensure the proper folders are synced? [Hard]

- Tableau Server HA (High Availability) involves setting up multiple server nodes for redundancy.
- Folder synchronization ensures that workbooks and other resources are replicated across all HA nodes.

- Refer to the official Tableau documentation for detailed configuration steps on HA and folder synchronization specific to your Tableau Server version.

Q34. Can I take daily snapshots of Tableau Server's database to analyze workbook usage (views over time)? [Hard]

Taking daily snapshots of the Tableau Server database for workbook usage, user logins, and data source usage requires advanced configurations. This likely involves Tableau Server APIs or scripting and would require working with a Tableau Server administrator.

Q35. How do I trigger an extract based off a database table being refreshed, instead of relying on Tableau's schedules? [Hard]

Tableau Server's built-in scheduling doesn't directly trigger extracts based on database table refreshes.

Possible workarounds:

- **Database Triggers:** If your database supports triggers, create a trigger that fires upon table refresh. This trigger can then initiate a Tableau Server extract refresh through the REST API (requires scripting).
- **External Monitoring:** Set up external monitoring tools to detect database table updates. These tools can then trigger a Tableau Server extract refresh script via the REST API.

Q36. Can I automatically add users via the REST API instead of manually adding them? [Hard]

Yes, you can automate user addition using Tableau Server's REST API. This eliminates manual user creation.

- **Familiarize yourself with the REST API:** refer to Tableau's documentation for user management endpoints and authentication methods.
- **Develop a script:** create a script (e.g., Python, Bash) that interacts with the Tableau Server REST API to add users, providing user information (username, email, etc.) and permissions.
- **Schedule the script:** schedule it to run periodically (e.g., daily) to add new users as needed.

Q37. How can I schedule automatic backups and sync the .tsbak file with AWS? [Hard]

Tableau Server offers built-in backup functionality — schedule automatic backups in the **Settings** → **Backup** section.

Syncing backups with AWS:

- Configure Tableau Server backup to a network share accessible by Tableau Server and AWS S3.
- Create a script that copies the backup files from the network share to your AWS S3 bucket after each scheduled backup.
- Schedule the script to run after each Tableau Server backup for timely synchronization with S3.

Q38. How can I find out what users/groups have what permissions on my workbooks? [Medium]

You can view user and group permissions on workbooks through the Tableau Server web interface:

- Log in to Tableau Server as an administrator.
- Navigate to the specific workbook.
- Click on the **Share** icon (or Permissions).
- In the permissions section, you'll see a list of users and groups with their assigned permissions (View, Edit, etc.) for that workbook.

Q39. Scenario: On Monday you create a worksheet with a filter and a parameter both based on Employee_ID. Monday evening, 6 new employees are inserted. Tuesday morning the data source is refreshed. How many distinct entries in the filter vs the parameter, and why? [Hard]

Employee ID Filter:

- **Specific IDs selected:** If you selected specific employee IDs (e.g., 1, 2, 3) in the filter on Monday, it will still only show those same IDs (3) on Tuesday. The new employees won't be included unless you manually adjust the filter.
- **All employee IDs selected:** If you selected all employee IDs on Monday, it will continue to show all IDs (including the new ones) on Tuesday — 10 distinct entries (original 4 + 6 new).

Employee ID Parameter: Regardless of the filter selection, the parameter will always reflect **all** employees in the table after the data refresh — **10 distinct entries** on Tuesday morning.

Explanation: The filter controls which data is displayed based on your selection and remains static unless manually changed. The parameter is a dynamic list that populates its values from the underlying data source, so it automatically updates when the data refreshes.

Q40. Calculating time between dates in the SAME column (e.g. time between orders) [Hard]

Tableau easily calculates time between two dates in *different* columns: `DATEDIFF('day', [start date], [end date])`. But order data often stores dates in a single column — how do we calculate time between consecutive orders?

Quick solution — LOOKUP: use the LOOKUP table calculation to compare each row with the previous one. This works, but table calculations run in-memory and can hamper performance on larger data sets.

Better solution — Custom SQL with DENSE_RANK:

- Connect to the data, drag in the orders table, and choose "Convert to Custom SQL".
- Use a `DENSE_RANK()` command to rank order dates (ORDER BY), restarting for every customer (PARTITION BY) — this numbers each customer's orders 1 through N.
- Add a second Custom SQL table with the same query, but change the DENSE_RANK to `+1` and name it "Previous Order Number" (rename the date as "Previous Order Date"). The +1 shifts the data set down by one row.
- Join on "Order Number = Previous Order Number" and "Customer Name = Customer Name".

Now consecutive dates sit side by side and a simple DATEDIFF gives time between orders — and it performs well even on big data, e.g. average days between orders as a bar chart per region.

More asked questions

Q41. What are the dimensions and facts? [Easy]

Dimensions are descriptive attributes that categorize data (customer name, product category, region, date) — the who/what/when/where. **Facts (measures)** are the quantitative values you analyze (sales amount, units sold, profit) — the how much/how many. Combined, they answer questions like "sales by region by month."

Q42. What type of data sources did you work on while creating dashboards? [Easy]

Relational databases (SQL Server, MySQL, PostgreSQL), spreadsheets (Excel/CSV), cloud sources (Google Sheets, Salesforce, Google Analytics), and flat files (JSON, text). Mention only the ones you can defend with follow-up answers.

Q43. Did you create dashboards from a data warehouse as a source? [Easy]

If yes: "Yes — we used [Snowflake/Redshift/SQL Server DW] as a centralized, integrated source optimized for analytics; I connected to fact and dimension tables and mostly used extracts on top for dashboard speed." If no, say you've worked with the same star-schema concepts on regular databases.

Q44. How can I archive valuable workbooks and have version control? [Medium]

Tableau Server keeps **revision history** (restore previous versions from the workbook's menu). For true archiving: download .twbx snapshots on a schedule (tabcmd/REST API) to a versioned store like Git or S3.

Q45. How can I be notified immediately when my extract fails or succeeds? [Medium]

Enable email notifications: Settings → Jobs → Email Notifications on Tableau Server, or account settings → "extract refresh failure" emails. For richer alerting, poll job status via the REST API and push to Slack/phone with a small Python script.

Q46. Can I remove older workbooks and simultaneously email the owner? [Hard]

Not built-in as one action. Do it manually (remove + email), or automate with the **REST API**: query stale workbooks (last accessed date from the repository), download an archive copy, delete, and send an email — all in one script.

Q47. What calculations are in my workbooks and how do I grab the formula? [Medium]

In Desktop: right-click the calculated field in the Data pane → **Edit** to see the formula. For an inventory across many workbooks, tools that parse the workbook XML (a .twb is XML) can list every calculation automatically.

Q48–50. Daily snapshots of Tableau Server's database for workbook usage / user logins / data source usage? [Hard]

Yes — the Tableau Server **repository (workgroup PostgreSQL)** holds views like historical_events, http_requests and views_stats. Enable repository access, snapshot the needed tables daily (or connect Tableau to the repository itself), and you can trend views per workbook, logins per user and data source hits over time.

Q51. How do I add a maintenance message to notify users of a server restart/upgrade? [Medium]

Use the server's **sign-in customization / welcome banner** (Site Settings) for a notice, or send a broadcast email to all users via groups. Larger orgs put a banner via custom portal pages where dashboards are embedded.

Q52. Can a parameter be used inside another parameter? [Medium]

No — a parameter is a standalone scalar input; parameters can't reference other parameters directly. You combine them inside a **calculated field** instead (e.g. one calc using two parameters), which achieves the same effect.

Q53. Can we use groups in calculated fields? [Medium]

Yes — a group appears as a field and can be referenced in calculations, e.g. `IF [Region Group] = "North Zone" THEN ...`. (In very old versions this was limited; modern Tableau supports it.)

Q54. Can we group multiple dimensions using groups? [Medium]

A group is built on **one dimension's members**. To group across multiple dimensions, first create a **combined field** (select both dimensions → Create → Combined Field) and group that, or use a calculated field / set with a multi-condition formula.

Q55. How many types of extensions are there in Tableau and how are they used? [Medium]

Main file types: `.twb` (workbook), `.twbx` (packaged workbook), `.hyper`/`.tde` (extract), `.tds`/`.tdsx` (data source / packaged), `.tbm` (bookmark), `.tps` (preferences/color palettes), `.tfl`/`.tflx` (Prep flows). Also **dashboard extensions** — web add-ins (.trex) that add functionality like write-back inside dashboards.

Q56. Live-connection dashboard published to server — how does the server stay connected to the database? [Medium]

The published workbook stores the **connection details with embedded credentials**. Every time a user opens or interacts with the view, VizQL Server sends live queries to the database, so data is always current — no refresh schedule needed (only the cache may briefly serve results, controllable via cache policies).

Q57. How many charts are there in Tableau and how many have you created? [Easy]

Show Me offers **24 chart types**; beyond that customs like waterfall, donut, funnel, Pareto are built manually. Safe answer: name 8–10 you genuinely use (bar, line, area, pie/donut, scatter, map, heat map, treemap, Gantt, bullet) and one custom chart you can explain step by step.

Q58. A country/province is missing and shows null in map view. What do you do? [Medium]

Click the "unknown" indicator (bottom-right) → **Edit Locations** → match the unrecognized names to the right geography (fix spellings like "Orissa" → "Odisha"), set the correct geographic role, or provide custom latitude/longitude. Filter genuine nulls if the row has no location at all.

Q59. Find the customer with the lowest overall profit. What is their profit ratio? [Medium]

Build: Customer on Rows, SUM(Profit) sorted ascending — the top row is the lowest-profit customer. Profit ratio = `SUM([Profit]) / SUM([Sales])` as a calculated field, formatted as %. (In Superstore data this classically lands on a heavily-discounted customer with a negative ratio.)

Q60. How do you handle nulls and other special values? [Medium]

Options: the null indicator's **"Filter data"** or **"Show data at default position"**; replace with `ZN([Measure])` (null→0) or `IFNULL([Field], "Unknown")`; fix at source in Prep/Power Query when it's a data-quality issue. Choose based on whether null means zero, unknown, or bad data.

Q61. Difference between sets and groups? Examples. [Medium]

Group: a static combination of members into higher-level buckets (East+West → "Coasts") — no logic, no in/out. **Set:** a dynamic subset based on a condition (Top 10 customers by sales — membership updates with data), supports IN/OUT comparison and combined sets. Rule: fixed labelling → group; conditional/dynamic membership → set.

Q62. Display top five and last five sales in the same view? [Hard]

Approach: create two sets on the same field — Top 5 by SUM(Sales) and Bottom 5 (Top 5 "by lowest") → **combine the sets** (Create Combined Set, "shared members in either") → filter the view on the combined set. Alternative: a rank calculation `RANK(SUM([Sales]))` filtered to rank ≤ 5 OR rank ≥ total – 4.

Filters, extracts & blending bank

Q1. What are the different types of filters in Tableau? Purpose of each filter? *[Easy]*

In order of execution: **Extract filters** (limit data while creating an extract), **Data source filters** (restrict data at connection level for all sheets), **Context filters** (create a temporary subset that other filters run on), **Dimension filters** (filter categorical values), **Measure filters** (filter on aggregated values), and **Table calculation filters** (applied last, after calculations, so they hide marks without changing underlying results).

Q2. What is the order of execution of filters in Tableau? *[Easy]*

Extract filters → Data source filters → Context filters → Filters on dimensions → Filters on measures → Table calculation filters. **Memorize this sequence** — it is one of the most repeated Tableau interview questions.

Q3. Have you ever used a context filter? When and why? Pros and cons? *[Medium]*

Yes — for example to show the **Top 10 products within a selected region**. Region must be a context filter, otherwise Top 10 is computed on all regions first. **Pros:** forces filter priority, can improve performance by shrinking the working dataset. **Cons:** the temporary context table must be recomputed whenever the filter changes, which can be slow on large data; too many context filters hurt performance.

Q4. Difference between a normal/cascading filter and a context filter? *[Medium]*

A **normal filter** works independently on the full dataset. A **cascading filter** ("Only relevant values") just changes what options are *visible* in another filter. A **context filter** actually changes the *data* other filters and calculations operate on — Top N and FIXED LOD calculations respect context filters but ignore normal dimension filters.

Q5. What is a Data Source filter? Advantages and disadvantages? *[Medium]*

A filter applied at the data-source level that restricts data for **every sheet and user** of that source. **Advantages:** one place to enforce rules (e.g. only 2 years of data, or row-level security), better performance, consistent across workbooks. **Disadvantage:** analysts can't access excluded data even when they need it, and it's easy to forget it exists when debugging "missing" data.

Q6. How to create a cascading filter (Region → Country)? *[Easy]*

Add both filters to the view → on the Country filter dropdown choose **"Only relevant values"**. Now selecting a region limits the country list to that region's countries. For performance on big data, make Region a context filter as well.

Q7. How to apply a common Apply button for all filters at once? *[Medium]*

On each quick filter's dropdown, select **Customize** → **Show Apply Button**. Each filter then waits for Apply. For one combined apply across many filters, the standard approach is to convert filters to **parameters + calculated field**, or place filters in a container so users set all values before the dashboard queries (Tableau has no single built-in global Apply for multiple quick filters).

Q8. How to make a Reset button that resets all filters at once? *[Medium]*

Create a dashboard **Button object** → set its action to navigate to the same dashboard exported in its default state, or use a **"Reset" dashboard action**: save the default view as the published state, then the toolbar's "Revert" resets everything. Alternative: a parameter-driven approach or the newer **dashboard navigation button + "Revert"** combination.

Q9. How do you use the LOOKUP function with filters? Any impact? *[Hard]*

`LOOKUP(expression, offset)` is a table calculation — it runs *after* normal filters on the values visible in the view. So a dimension filter changes what LOOKUP sees and can break offsets. That's why "hiding" with a **table-calc filter** (e.g. LOOKUP(MIN([Date]),0) used as filter) is the trick to filter the view without disturbing LOOKUP/running calculations.

Q10. Will filters work when we do data blending? *[Medium]*

Yes, but with limits: filters apply to their own data source. A filter from the primary source doesn't automatically filter the secondary source — to filter across sources you must add the linking field to the view, use **filter actions**, or (in newer versions) **cross-data-source filters** on the common field.

Q11. How to get cross-data-source filters? Is it possible in Tableau? *[Medium]*

Yes — since Tableau 10, if two data sources share a field (same name/type or mapped via Edit Relationship), right-click the filter → **Apply to worksheets** → **All using related data sources**. The filter then filters both sources together.

Q12. Difference between Quick filter and Normal filter? *[Easy]*

They are the same filter underneath — a **quick filter** is just the interactive filter card *shown on the view/dashboard* so end users can change values, while a "normal" filter sits on the Filters shelf, set by the author. Note: many visible quick filters (especially "Only relevant values") slow dashboards because each renders its own query.

Q13. What is a User filter? How to use it? [Medium]

A user filter maps Tableau Server users/groups to allowed data values (Server → Create User Filter → pick field → assign members to values). When published, each user sees only their rows — the manual way to achieve row-level security. For scale, prefer a calculated field using `USERNAME()` against an entitlement column.

Q14. What is Row Level Security? Ways to achieve it in Tableau? [Hard]

RLS restricts which *rows* each user sees in the same workbook. Three ways: **1) Manual user filters** (quick, but high maintenance), **2) Calculated field with USERNAME()** — e.g. `USERNAME() = [Manager Email]` used as a data source filter, **3) Entitlement table** joined to the data and filtered on the logged-in user (enterprise standard). Always apply as a data source filter so it can't be removed sheet by sheet.

Q15. How do you check performance in Tableau? Any tool used? [Medium]

Use the built-in **Performance Recording** (Help → Settings and Performance → Start Performance Recording). Perform the slow actions, stop the recording, and Tableau opens a workbook showing time spent on query execution, layout and calculations. On Server, admin views and the `http_requests` repository data help too.

Q16. Explain the complete Performance Recording option. [Medium]

Start recording → interact with the workbook (open, filter, change parameters) → stop recording. Tableau generates a performance workbook with a timeline of events: **executing query, computing layout, geocoding, blending, computing table calcs**. Sort by duration to find the bottleneck — if "executing query" dominates, optimize the data source/extract; if "computing layout", reduce marks and visuals.

Q17. A published workbook takes 3–5 minutes to open and the user is unhappy. What is your first step? [Hard]

First step: run a performance recording to find out *where* the time goes — query, rendering or calculations. Then apply the matching fix: switch live → extract, aggregate data, reduce quick filters (especially "Only relevant values"), cut the number of marks/visuals per dashboard, or move heavy logic from table calcs into the extract/source.

Q18. What is meant by extract and how do you take an extract? [Easy]

An extract is a compressed snapshot of your data stored in Tableau's fast in-memory format (.hyper). To create: on the Data Source page select **Extract** (instead of Live) → optionally add extract filters/aggregation → click the sheet and save the extract file. Extracts make dashboards faster and work offline.

Q19. Where does the extract file get saved on the system? [Easy]

By default in the **My Tableau Repository** → **Datasources** folder of your user documents (as a .hyper file). When you save a .twbx packaged workbook, the extract is bundled inside the package.

Q20. What is the difference between Live and Extract connection? [Easy]

Live: queries the source database in real time — always fresh, but speed depends on the database. **Extract:** a local snapshot in Tableau's engine — much faster for dashboards, works offline, but shows data only as of the last refresh.

Q21. Live or Extract — which performs better and when would you choose each? [Easy]

Extract usually performs better because Tableau's Hyper engine is optimized for analytics. Choose **Live** when data must be real-time (stock levels, operations monitoring) or the database is a fast warehouse; choose **Extract** for typical reporting, big aggregations, slow sources, or offline sharing.

Q22. How to schedule an extract in Tableau? [Easy]

Publish the workbook/data source with embedded credentials → on Tableau Server/Cloud open it → **Scheduled Tasks / Refresh Schedules** → **Add a schedule** (e.g. daily 6 AM). The Backgrounder process runs the refresh automatically.

Q23. Can we do an incremental refresh in Tableau? How? [Medium]

Yes. In the Extract dialog choose **Incremental refresh** and pick a column that identifies new rows (usually a date or increasing ID). Tableau appends only new rows on each refresh. Do a periodic **full refresh** too, because incremental won't capture updates or deletes to existing rows.

Q24. What is the use of aggregation when creating an extract? [Medium]

"Aggregate data for visible dimensions" rolls the extract up to the level of detail you actually use (e.g. daily totals instead of row-level transactions). The extract becomes dramatically smaller and faster — ideal when you never need transaction-level drill-down.

Q25. What is meant by data blending? How do you do it and what is it for? [Medium]

Blending combines **different data sources** in one view: build the view from the primary source, then drag a field from the secondary source — Tableau links them on the common field (orange link icon). Each source is aggregated separately and merged, like a left join from the primary. Use it when sources can't be joined directly (e.g. Excel targets + database actuals).

Q26. Can we join Excel and another data source together in Tableau? [Easy]

Yes — via **cross-database joins**: in the same data source click Add → connect Excel and the database → drag both tables to the canvas and define the join. If a direct join isn't possible, use data blending instead.

Q27. How do you join two different data sources? [Medium]

Two options: **1) Cross-database join** — add both connections inside one Tableau data source and join at row level; **2) Data blending** — keep them as separate sources and blend on a common field in the view. Join = row level before aggregation; blend = after aggregation.

Q28. What type of join does data blending perform? Difference between join and blending? [Medium]

Blending behaves like a **left join from the primary source** on aggregated results. Differences: a join combines row-level data from the *same* data source (or cross-database) *before* aggregation; a blend combines *separately aggregated* results from *different* sources. Row-level detail from the secondary source is not available in a blend.

Q29. What problems have you faced during data blending? [Hard]

Common real issues: the *** (asterisk)** appearing when multiple secondary values match one primary row; nulls when linking fields don't match exactly (spelling/case/data type); secondary fields can't be used on certain shelves; COUNTD and some calculations not supported from the secondary source; and performance drops when blending on high-cardinality fields.

*Q30. Why do you see a * (asterisk) after blending? [Medium]*

The asterisk means **multiple values from the secondary source matched a single row of the primary source** — the blend can't show them all in one cell, so it shows *. Fix by blending at the correct level of detail (add the missing linking field) or aggregating the secondary source first.

Q31. How do you create relationships between two data sources for blending? [Medium]

Tableau auto-links fields with identical names and types. For different names go to **Data** → **Edit Blend Relationships** → Custom → map the fields (e.g. "Cust ID" ↔ "Customer ID"). In the view, click the link icon next to the field to activate/deactivate blending on it.

Q32. How many charts are there in Tableau? [Easy]

The **Show Me panel offers 24 chart types** (bar, line, area, pie, map, scatter, treemap, bullet, box-and-whisker, Gantt, histogram, heat map, highlight table, etc.). Beyond Show Me you can build custom charts — waterfall, donut, funnel, Pareto, Sankey — using dual axes and calculations.

Q33. Have you built a chart apart from the default ones? What and how? [Medium]

Good answer example: "Yes, a **donut chart** — a pie chart on a dual axis where the second (smaller, blank) pie creates the hole; and a **waterfall chart** — a Gantt bar on running total of the measure with negative size." Pick one you can actually reproduce and explain step by step.

Q34. Two measures — profit and sales by country, with filters. Which chart do you prefer? [Medium]

Either a **dual-axis combo** (bars for sales, line for profit) or a **scatter plot** (sales vs profit, one mark per country) — the scatter instantly exposes high-sales-but-low-profit countries. Add a region filter with "Only relevant values" for interactivity, and justify whichever you choose.

Q35. What is the use of a scatter plot? When do you use it? [Easy]

A scatter plot shows the **relationship between two numeric measures** — each mark is one entity. Use it to spot correlation, clusters and outliers, e.g. discount vs profit by sub-category to see where heavy discounting kills margin. Add trend lines to quantify the relationship.

Q36. Know all the charts and the scenarios they're useful for. *[Medium]*

Quick map: **Bar** — compare categories; **Line** — trends over time; **Pie/Donut** — parts of a whole (few slices); **Scatter** — relationship of two measures; **Heat map** — intensity across a matrix; **Treemap** — hierarchical share; **Histogram** — distribution; **Box plot** — spread and outliers; **Gantt** — duration/schedules; **Bullet** — actual vs target; **Map** — geographic patterns; **Waterfall** — how increments build a total.

Q37. Have you built a pie or donut chart? Explain how (donut). *[Medium]*

Donut steps: create a pie of the measure by category → drag the same aggregated measure to Rows twice → make it dual axis → on the second axis remove color/labels and shrink its size, set color to background → this hollow circle over the pie forms the donut → add a total label in the centre.

Q38. What is the difference between discrete and continuous fields? *[Easy]*

Discrete (blue): distinct labels/headers — creates separate panes/headers (e.g. Year as a label). **Continuous (green):** unbroken range — creates an axis (e.g. Sales axis, exact date axis). The same field can often be used either way; it changes how Tableau draws the view.

Q39. What are LOD expressions? Explain FIXED, INCLUDE, EXCLUDE with example. *[Hard]*

Level of Detail expressions compute values at a level independent of the view. **FIXED** — exact level: `{FIXED [Customer]: SUM([Sales])}` gives each customer's lifetime sales on every row. **INCLUDE** — adds a dimension to the view's level (e.g. average of per-order sales). **EXCLUDE** — removes a dimension (e.g. show regional total next to each city). Classic use: cohort analysis with `{FIXED [Customer]: MIN([Order Date])}` for first-purchase date.

Q40. What are parameters and how are they different from filters? *[Medium]*

A **parameter** is a single global input value (number/string/date) that users control; by itself it does nothing until used in a calculation, filter, reference line or Top N. A **filter** directly restricts data. Parameters can do what filters can't: switch measures dynamically, drive What-If values, work across data sources.

Tableau Server bank

Q1. What is Tableau Server? *[Easy]*

A platform for publishing and sharing interactive dashboards within an organization — it handles user access, permissions, scheduled refreshes, collaboration and governance. Deployed on company infrastructure (or use Tableau Cloud as the hosted version).

Q2. Difference between Tableau Server and Tableau Online (Cloud)? *[Easy]*

Server: installed and managed on your own infrastructure — full control, works inside private networks, you handle upgrades and backups. **Online/Cloud:** SaaS hosted by Tableau — no infrastructure to manage, faster to start, accessible anywhere.

Q3. How to publish a workbook to Tableau Server? *[Easy]*

In Tableau Desktop: **Server** → **Publish Workbook** → sign in → choose the project → set the name, tags, permissions, refresh schedule and credentials → Publish. The workbook is then available in the browser to permitted users.

Q4. What is a project in Tableau Server? *[Easy]*

A project is a **folder on the server** that organizes workbooks and data sources and is the main unit for permissions — e.g. a "Finance" project where only the finance group has access. Projects can contain sub-projects for finer structure.

Q5. What are the components available in Tableau Server? *[Medium]*

Sites (isolated tenants), **projects**, **groups and users**, **workbooks and data sources**, **schedules** (extract refresh, subscriptions), **permissions**, plus admin areas for monitoring, performance tuning and server status.

Q6. How do you give permissions to a workbook to a user? *[Easy]*

Open the workbook on the server → **Permissions** → add the user or group → choose a template (View, Explore, Publish, Administer) or set capabilities individually (view, filter, download, web edit, delete). Best practice: assign permissions to **groups at the project level**, not per user per workbook.

Q7. How many levels of security are there in Tableau Server? [Medium]

Four practical layers: **1) Authentication** (who can log in — AD/SAML/local), **2) Object/content permissions** (site → project → workbook/view level), **3) Data source security** (connection credentials), **4) Row-level security** (which rows each user sees inside a view).

Q8. What are the different permission roles in Tableau Server? [Medium]

Site roles: **Server/Site Administrator, Creator, Explorer (can publish), Explorer, Viewer**. Content permission templates: View, Explore, Publish, Administer. Site role sets a user's maximum capability; content permissions decide what they can do on specific items.

Q9. How do you take a backup from Tableau Server? [Medium]

Using the TSM command line: `tsm maintenance backup -f backup-file.tsbak -d` — this backs up the repository and file store. Schedule it regularly and store the .tsbak off the server (e.g. sync to S3).

Q10. What are the different server processes in Tableau? [Hard]

Key processes: **Gateway** (routes requests), **Application Server (VizPortal)** — logins and browsing, **VizQL Server** — renders views and runs queries, **Backgrounder** — extract refreshes and subscriptions, **Data Server** — manages published data sources, **Repository (PostgreSQL)** — metadata, **File Store / Data Engine (Hyper)** — extract storage and query engine, **Cache Server**.

Q11. How do Tableau permissions/security work in order after user login? [Hard]

Order of evaluation: **authentication** → **site role** (maximum capability) → **content permissions** (deny beats allow; project → workbook → view) → **data security** (credentials + row-level security). Effective permission = most restrictive combination of these layers.

Q12. What if the server goes down suddenly? First step? [Hard]

First check server status: `tsm status -v` to see which process failed, and review logs. Communicate downtime to users, restart the failed service (`tsm restart`), and if hardware/corruption is involved, restore from the latest .tsbak backup. Root-cause afterwards (disk full and memory are the usual culprits).

Q13. How can we manage alerts when the server goes down? [Medium]

Enable built-in **email alerts** in TSM (server health notifications), monitor with external tools (Nagios, Datadog, CloudWatch) hitting Tableau's status endpoint, and set up the Backgrounder failure notifications so admins are emailed automatically.

Q14. Where is the server installed? Basic requirements/configuration? [Medium]

On a dedicated Windows/Linux machine (on-prem or cloud VM). Baseline for production: **8+ cores, 32GB+ RAM, 500GB+ SSD**, 64-bit OS. Bigger deployments scale to multi-node clusters with roles split across nodes.

Q15. Have you done the installation process for a server? How? [Medium]

Outline answer: download installer → run setup → initialize TSM → activate license → configure identity store (local/AD), gateway port and run-as account → initialize server → create the first admin → then configure sites, SMTP for emails, and backup schedule.

Q16. User has editor permission on a project but can't view data in a shared workbook. How do you handle it? [Hard]

Diagnose layer by layer: 1) workbook-level permission may **deny** view (deny overrides project allow), 2) the **data source credentials** may not be embedded — the user is prompted or blocked, 3) **row-level security** may filter out all their rows. Check effective permissions on the workbook, embed credentials or fix the RLS mapping accordingly.

Q17. Have you done ad-hoc reporting in Tableau? How is it achieved? [Medium]

Yes — via **web edit** on Tableau Server (Explorers open a published workbook and modify/answer questions in the browser) and by publishing **certified data sources** so business users build their own views with "Ask Data"/self-service without touching Desktop.

Q18. Is it possible to do development in Tableau Server? [Easy]

Yes, limited development via **web authoring (web edit)** — create and edit workbooks in the browser against published data sources. Full-featured development (complex joins, some advanced features) still happens in Tableau Desktop.

Q19. What does the Edit option for a workbook on the server do? ***[Easy]***

It opens the workbook in **web authoring mode** — the user can modify sheets, add calculations and save (or Save As) directly in the browser, without needing Tableau Desktop, if their role and permissions allow web edit.

Q20. Difference between doing development on Server vs Desktop? ***[Easy]***

Desktop: full feature set — all connectors, complex data prep, extracts, faster iteration offline. **Server web edit:** convenient, no install, works on published sources, but limited data-source editing and some features missing. Typical flow: build in Desktop → publish → tweak on Server.

Q21. Can we publish data sources to Tableau Server? ***[Easy]***

Yes — **Server** → **Publish Data Source**. A published (and certified) data source becomes a single governed source of truth: many workbooks connect to it, refreshes are scheduled once, and calculations/aliases are shared.

Q22. Live connection to a local Excel published to server — will changes in the local Excel reflect after refresh? Why? ***[Hard]***

No. When you publish a workbook with a live connection to a *local* file, Tableau packages a copy of that file to the server. The server refreshes against its own copy — it cannot see your local machine. Changes reflect only if the file sits on a network/cloud path the server can reach (UNC path, SharePoint/OneDrive), or if you republish.

Q23. Is performance monitoring possible on the server? How? ***[Medium]***

Yes — built-in **Administrative views** (Status → traffic, background tasks, slow views, disk usage), performance recording on individual views, and querying the **Tableau Server repository** (workgroup PostgreSQL) for deeper custom analysis.

Q24. Have you faced a timeout error? How do you handle it? ***[Hard]***

Yes — long queries hitting the default limits. Fixes: optimize the query/extract (the real cure), or raise limits: `vizqlserver.querylimit` (query timeout) and session timeouts via TSM. Also check the database side timeout. Always prefer making the view faster over raising the timeout.

Q25. How do you schedule a normal workbook with a live connection? ***[Medium]***

Live connections don't need data refreshes — the data is always current. What you can schedule for them is **subscriptions** (emailed snapshots on a schedule) and **cache refresh policies**. Refresh schedules apply to extracts, not live connections.

Q26. How do you automate reports using Tableau? ***[Medium]***

Combine: **extract refresh schedules** (data stays current) + **subscriptions** (dashboards emailed to stakeholders daily/weekly) + **data-driven alerts** (email when a metric crosses a threshold) + the **REST API/tabcmd** for custom automation like PDF export to a shared drive.

Q27. What platforms can Tableau Server run on? ***[Easy]***

Windows Server and **Linux** (RHEL, CentOS/Rocky, Ubuntu, Amazon Linux), on-premises or on cloud VMs (AWS, Azure, GCP). Tableau Cloud is the option with no OS to manage at all.

Q28. Can one workbook connect to different sites/instances of the same server as different data sources? ***[Hard]***

Published data sources are scoped to a **single site** — a workbook lives in one site and can use that site's published sources. To combine data across sites, connect directly to the underlying databases instead, since cross-site references aren't supported.

Q29. Is it possible to change user authentication within one installation? ***[Hard]***

You can switch the identity store type (local ↔ Active Directory) but it's **disruptive** — typically requires export/import of content and recreating users. Per-site you can vary some auth options (e.g. SAML on specific sites). Plan authentication before going live.

Q30. How to handle giving permissions to multiple users? ***[Easy]***

Create **groups** (or sync from Active Directory) → assign permissions to groups at the **project level** → lock content permissions to the project. New users just get added to a group and inherit everything — never manage permissions user by user.

Q31. What challenges have you faced in Tableau Server? [Medium]

Real-world examples to mention: extract refresh failures due to expired credentials, slow dashboards at peak load (fixed via extracts and fewer quick filters), permission conflicts from deny rules, disk filling with old extracts, and coordinating downtime for upgrades.

Q32. Have you embedded a Tableau URL with other applications? [Medium]

Yes — dashboards can be embedded in web apps/SharePoint/Salesforce via the **Embedding API / iframe share link**, with URL parameters or JS API for filtering (e.g. `?Region=West`), and trusted authentication or connected apps for single sign-on.

Q33. Explain Tableau Server architecture. [Hard]

A multi-tier architecture: **Gateway** receives requests → **Application Server** handles login/browsing → **VizQL Server** turns interactions into queries and renders views → **Data Server/Data Engine (Hyper)** serve data and extracts → **Backgrounder** runs refreshes/subscriptions → **Repository (PostgreSQL)** stores metadata → **File Store** holds extracts. All processes can be distributed across nodes for scale and HA.

Q34. A published workbook URL is given to a user — what decides whether they can open it? [Medium]

The chain: valid **login** to that site → a site role of at least Viewer → **View permission** on the workbook (no deny anywhere up the chain) → access to the data (embedded credentials or their own) → RLS then decides which rows they see.